1.0 Introduction

With the advent of data-driven decision-making, information analytics has become very important in converting raw data into valuable insights that can be used to improve organizational strategies and operations. This capability of cleaning, reducing, transforming, and analyzing data is essential to ensure the accuracy, reliability, and usefulness of analytical findings. The low quality of the data or an incorrect method of analysis can result in false conclusions and decision-making.

The dataset used in this study is Global Superstore Orders, which is based on the initial “global_superstore_2016.xlsx” dataset that was acquired on Kaggle. The dataset comprises 24 attributes and 5,000 records, characterising various facets of an international retail company, including segment, sales, profit, shipping cost, order priority, discount, product name, and shipping mode. The dataset is rather comprehensive, which contributes to its appropriateness in the implementation of major ideas and methods in information analytics.

This study aims to use analytical and systematic data pre-processing techniques in order to obtain meaningful information about the dataset. The tasks involve determining data quality problems by data cleaning, decreasing the complexity of data by using data reduction methods, and converting the chosen variables to the correct scale of measurement. Moreover, correlation analysis is done to determine the correlation between the chosen variables, such as “Quantity” and “Discount”, and “Shipping Cost” and “Order Priority”. Finally, the use of data visualization techniques is used to visually depict the major categorical variables of “Ship Mode”, “Segment”, “Category”, “Sub-Category”, and “Product Name”, which allows better interpretation and explanation of the results of the analysis.

2.0 Data Cleaning

2.1 Inconsistency: Data Format Standardisation

Problem Identification:

In the original dataset, the Order Date and Ship Date attributes were displayed in different system-dependent date formats, which could cause confusion and errors during sorting, filtering, and time-based analysis. Inconsistent date representations may also lead to misinterpretation when the dataset is processed using different software tools.

Data Cleaning Action:

To address this issue, all date values in the Order Date and Ship Date columns were standardised into a single, consistent format.

Standardised Format Applied:

DD-MMM-YYYY (eg. 19-Sep-2015)

Method Applied:

i. The Order Date and Ship Date columns were selected
ii. Format Cells → Date was applied
iii. The date format was standardised to DD-MMM-YYYY

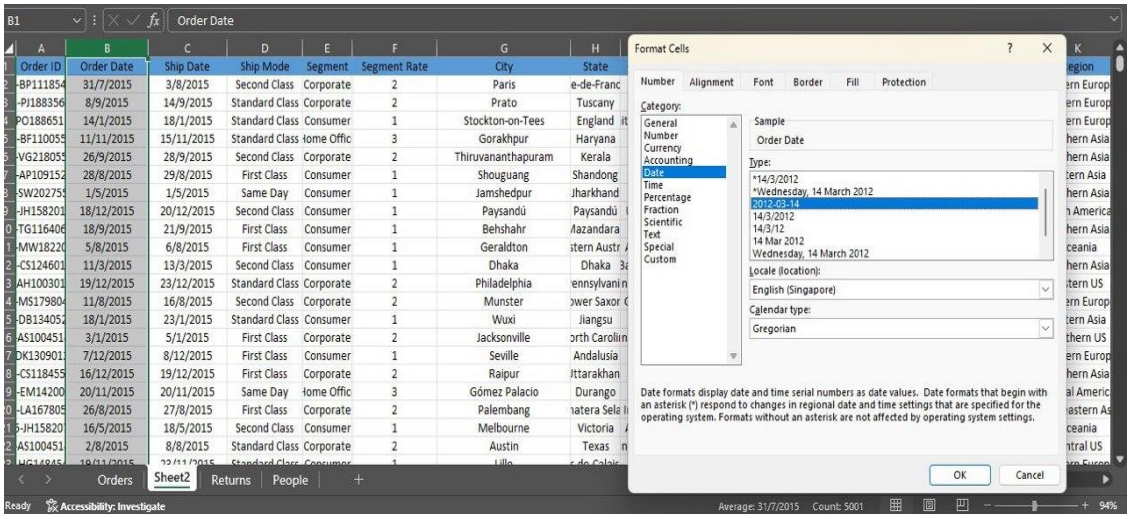


Figure 1: Order Date

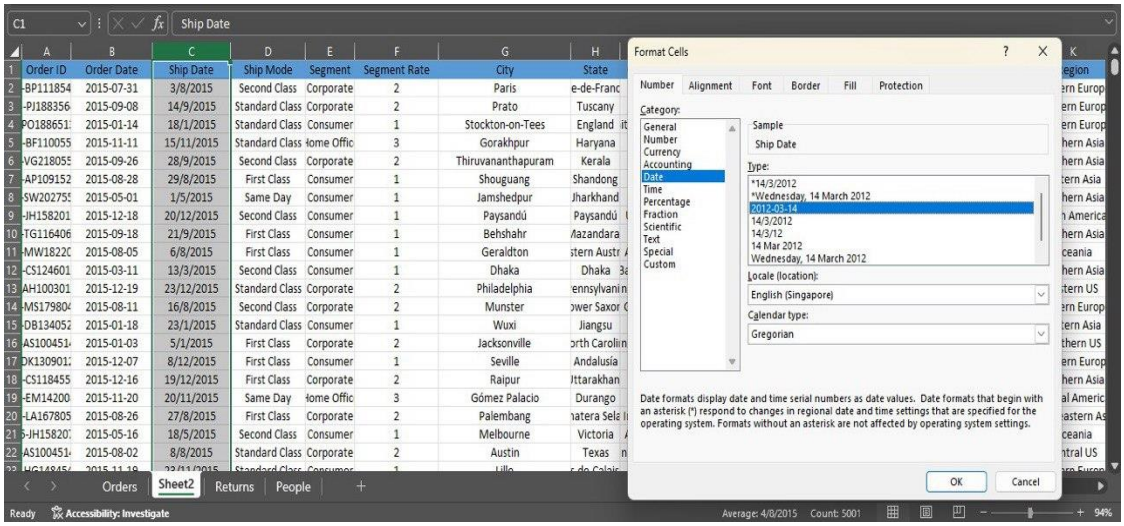


Figure 2: Ship Date

2.2 Incomplete Data: Missing Values

Problem Identification:

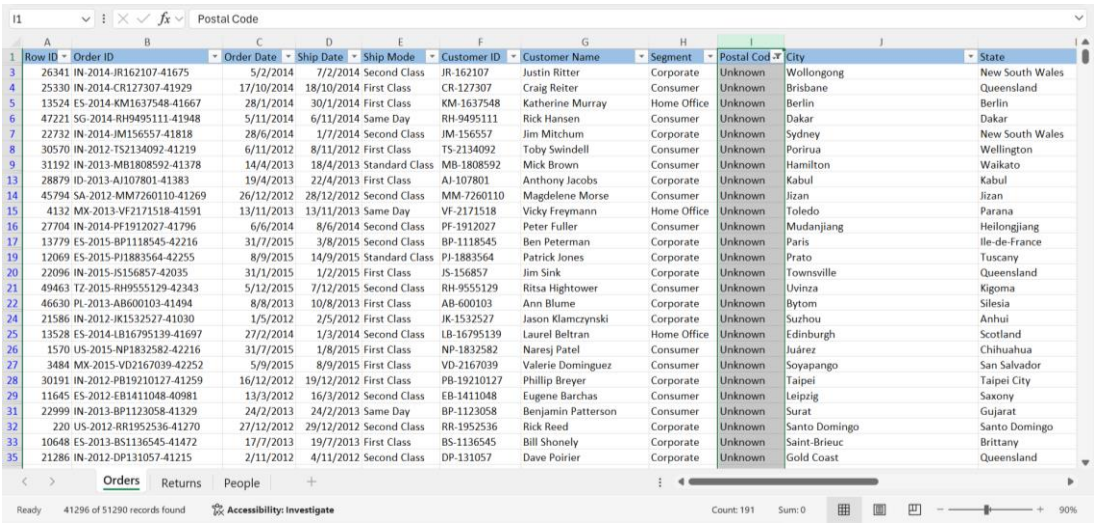
During data inspection, several records in the dataset were identified as containing missing or blank values in the Postal Code attribute. Incomplete data can impact data consistency and may lead to issues during grouping or filtering processes.

Data Cleaning Action:

To handle this issue, all missing Postal Code values were replaced with a standard placeholder value, namely “Unknown”. This approach was selected to preserve all transaction records while ensuring that no attribute contained blank values.

Method Applied:

i. The Postal Code column was filtered to identify blank cells
ii. Blank values were replaced with the value “Unknown.”
iii. The dataset was reviewed to confirm that no missing values remained



Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Postal Cod	City	State
26341	IN-2014-JR162107-41675	5/2/2014	7/2/2014	Second Class	JR-162107	Justin Ritter	Corporate	Unknown	Wollongong	New South Wales
25330	IN-2014-CR127307-41929	17/10/2014	18/10/2014	First Class	CR-127307	Craig Reiter	Consumer	Unknown	Brisbane	Queensland
13524	ES-2014-KM1637548-41667	28/1/2014	30/1/2014	First Class	KM-1637548	Katherine Murray	Home Office	Unknown	Berlin	Berlin
47221	SG-2014-RH9495111-41948	5/11/2014	6/11/2014	Same Day	RH-9495111	Rick Hansen	Consumer	Unknown	Dakar	Dakar
22732	IN-2014-JM156557-41818	28/6/2014	1/7/2014	Second Class	JM-156557	Jim Mitchum	Corporate	Unknown	Sydney	New South Wales
30570	IN-2012-TS2134092-41219	6/11/2012	8/11/2012	First Class	TS-2134092	Toby Swindell	Consumer	Unknown	Porirua	Wellington
31192	IN-2013-MB1808592-41378	14/4/2013	18/4/2013	Standard Class	MB-1808592	Mick Brown	Consumer	Unknown	Hamilton	Waikato
28879	ID-2013-AJ107801-41383	19/4/2013	22/4/2013	First Class	AJ-107801	Anthony Jacobs	Corporate	Unknown	Kabul	Kabul
45794	SA-2012-MM7260110-41269	26/12/2012	28/12/2012	Second Class	MM-7260110	Magdelene Morse	Consumer	Unknown	Jizan	Jizan
4132	MX-2013-VF2171518-41591	13/11/2013	13/11/2013	Same Day	VF-2171518	Vicky Freymann	Home Office	Unknown	Toledo	Parana
27704	IN-2014-PF1912027-41796	6/6/2014	8/6/2014	Second Class	PF-1912027	Peter Fuller	Consumer	Unknown	Mudanjiang	Heilongjiang
13779	ES-2015-BP1118545-42216	31/7/2015	3/8/2015	Second Class	BP-1118545	Ben Peterman	Corporate	Unknown	Paris	Ile-de-France
12069	ES-2015-PJ1883564-42255	8/9/2015	14/9/2015	Standard Class	PJ-1883564	Patrick Jones	Corporate	Unknown	Prato	Tuscany
22096	IN-2015-JS156857-42035	31/1/2015	1/2/2015	First Class	JS-156857	Jim Sink	Corporate	Unknown	Townsville	Queensland
49463	TZ-2015-RH9555129-42343	5/12/2015	7/12/2015	Second Class	RH-9555129	Ritsa Hightower	Consumer	Unknown	Uvinza	Kigoma
46630	PL-2013-AB600103-41494	8/8/2013	10/8/2013	First Class	AB-600103	Ann Blume	Corporate	Unknown	Bytom	Silesia
21586	IN-2012-JK1532527-41030	1/5/2012	2/5/2012	First Class	JK-1532527	Jason Klamczynski	Corporate	Unknown	Suzhou	Anhui
13528	ES-2014-LB16795139-41697	27/2/2014	1/3/2014	Second Class	LB-16795139	Laurel Beltran	Home Office	Unknown	Edinburgh	Scotland
1570	US-2015-NP1832582-42216	31/7/2015	1/8/2015	First Class	NP-1832582	Naresj Patel	Consumer	Unknown	Juárez	Chihuahua
3484	MX-2015-VD2167039-42252	5/9/2015	8/9/2015	First Class	VD-2167039	Valerie Dominguez	Consumer	Unknown	Soyapango	San Salvador
30191	IN-2012-PB19210127-41259	16/12/2012	19/12/2012	First Class	PB-19210127	Phillip Breyer	Corporate	Unknown	Taipei	Taipei City
11645	ES-2012-EB1411048-40981	13/3/2012	16/3/2012	Second Class	EB-1411048	Eugene Barchas	Consumer	Unknown	Leipzig	Saxony
22999	IN-2013-BP1123058-41329	24/2/2013	24/2/2013	Same Day	BP-1123058	Benjamin Patterson	Consumer	Unknown	Surat	Gujarat
220	US-2012-RR1952536-41270	27/12/2012	29/12/2012	Second Class	RR-1952536	Rick Reed	Corporate	Unknown	Santo Domingo	Santo Domingo
10648	ES-2013-BS1136545-41472	17/7/2013	19/7/2013	First Class	BS-1136545	Bill Shonely	Corporate	Unknown	Saint-Brieuc	Brittany
21286	IN-2012-DP131057-41215	2/11/2012	4/11/2012	Second Class	DP-131057	Dave Poirier	Corporate	Unknown	Gold Coast	Queensland

Figure 3: Missing values in the Postal Code column

2.3 Noisy Data: Duplicate Records

Problem Identification:

Duplicate records were found in the dataset, particularly repeated Order ID entries. Duplicate data can distort analysis results by overrepresenting certain transactions, leading to biased insights.

Data Cleaning Action:

Duplicate records were removed to ensure each order was uniquely represented in the dataset.

Method Applied:

i. The entire dataset was selected in Excel.

ii. The Remove Duplicates function was applied, focusing on the Order ID column.

iii. The dataset was reviewed to confirm that all duplicates were removed.

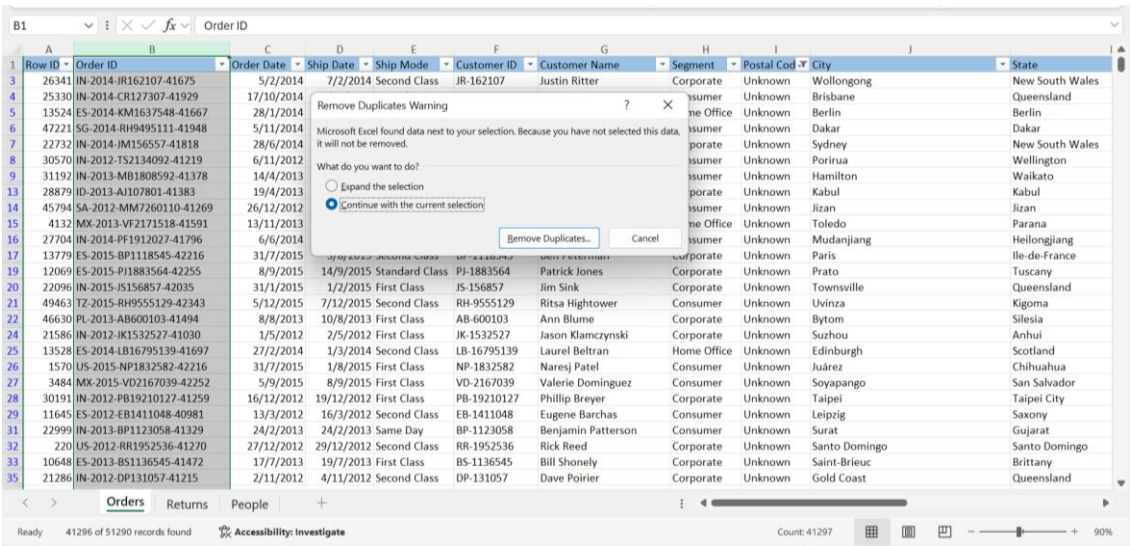


Figure 4: Duplicates Order ID

3.0 Data Reduction

3.1 Removing Unnecessary Column

Problem Identification:

The dataset contains several attributes that are not relevant to the analysis, such as Row ID, Customer Name, and Postal Code. Including these columns increases the dataset size and can slow down processing without adding value to the analysis.

Data Reduction Action:

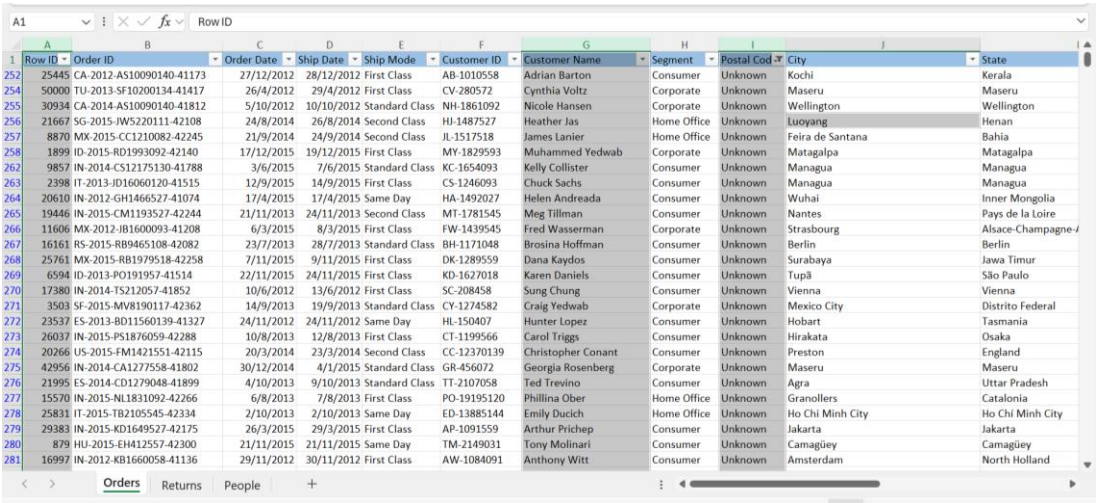
Unnecessary columns were removed from the dataset to reduce complexity and focus on meaningful attributes.

Method Applied:

i. The dataset was reviewed to identify irrelevant columns.

ii. Columns Row ID, Customer Name, and Postal Code were deleted in Excel.

iii. The dataset was verified to ensure only necessary attributes remained.



Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Postal Cod	City	State
25445	CA-2012-AS10090140-41173	27/12/2012	28/12/2012	First Class	AB-1010558	Adrian Barton	Consumer	Unknown	Kochi	Kerala
50000	TU-2013-SF10200134-41417	26/4/2012	29/4/2012	First Class	CV-280572	Cynthia Voltz	Corporate	Unknown	Maseru	Maseru
30934	CA-2014-AS10090140-41812	5/10/2012	10/10/2012	Standard Class	NH-1861092	Nicole Hansen	Corporate	Unknown	Wellington	Wellington
21667	SG-2015-JW5220111-42108	24/8/2014	26/8/2014	Second Class	HJ-1487527	Heather Jas	Home Office	Unknown	Luoyang	Henan
8870	MX-2015-CC1210082-42245	21/9/2014	24/9/2014	Second Class	JL-1517518	James Lanier	Home Office	Unknown	Feira de Santana	Bahia
1899	ID-2015-RD1993092-42140	17/12/2015	19/12/2015	First Class	MY-1829593	Muhammed Yedwab	Corporate	Unknown	Matagalpa	Matagalpa
9857	IN-2014-CS12175130-41788	3/6/2015	7/6/2015	Standard Class	KC-1654093	Kelly Collister	Consumer	Unknown	Managua	Managua
2398	IT-2013-JD16060120-41515	12/9/2015	14/9/2015	First Class	CS-1246093	Chuck Sachs	Consumer	Unknown	Managua	Managua
20610	IN-2012-GH1466527-41074	17/4/2015	17/4/2015	Same Day	HA-1492027	Helen Andreada	Consumer	Unknown	Wuhai	Inner Mongolia
19446	IN-2015-CM1193527-42244	21/11/2013	24/11/2013	Second Class	MT-1781545	Meg Tillman	Consumer	Unknown	Nantes	Pays de la Loire
11606	MX-2012-JB1600093-41208	6/3/2015	8/3/2015	First Class	FW-1439545	Fred Wasserman	Corporate	Unknown	Strasbourg	Alsace-Champagne-
16161	RS-2015-RB9465108-42082	23/7/2013	28/7/2013	Standard Class	BH-1171048	Brosina Hoffman	Consumer	Unknown	Berlin	Berlin
25761	MX-2015-RB1979518-42258	7/11/2015	9/11/2015	First Class	DK-1289559	Dana Kaydos	Consumer	Unknown	Surabaya	Jawa Timur
6594	ID-2013-PO1919574-41514	22/11/2015	24/11/2015	First Class	KD-1627018	Karen Daniels	Consumer	Unknown	Tupã	São Paulo
17380	IN-2014-TS212057-41852	10/6/2012	13/6/2012	First Class	SC-208458	Sung Chung	Consumer	Unknown	Vienna	Vienna
3503	SF-2015-MV8190117-42362	14/9/2013	19/9/2013	Standard Class	CY-1274582	Craig Yedwab	Corporate	Unknown	Mexico City	Distrito Federal
23537	ES-2013-BD11560139-41327	24/11/2012	24/11/2012	Same Day	HL-150407	Hunter Lopez	Consumer	Unknown	Hobart	Tasmania
26037	IN-2015-PS1876059-42288	10/8/2013	12/8/2013	First Class	CT-1199566	Carol Triggs	Consumer	Unknown	Hirakata	Osaka
20266	US-2015-FM1421551-42115	20/3/2014	23/3/2014	Second Class	CC-12370139	Christopher Conant	Consumer	Unknown	Preston	England
42956	IN-2014-CA1277558-41802	30/12/2014	4/1/2015	Standard Class	GR-456072	Georgia Rosenberg	Corporate	Unknown	Maseru	Maseru
21995	ES-2014-CD1279048-41899	4/10/2013	9/10/2013	Standard Class	TT-2107058	Ted Trevino	Consumer	Unknown	Agra	Uttar Pradesh
15570	IN-2015-NL1831092-42266	6/8/2013	7/8/2013	First Class	PO-19195120	Phillina Ober	Home Office	Unknown	Granollers	Catalonia
25831	IT-2015-TB2105545-42334	2/10/2013	2/10/2013	Same Day	ED-13885144	Emily Ducich	Home Office	Unknown	Ho Chi Minh City	Ho Chi Minh City
29383	IN-2015-KD1649527-42175	26/3/2015	29/3/2015	First Class	AP-1091559	Arthur Prichep	Consumer	Unknown	Jakarta	Jakarta
879	HU-2015-EH412557-42300	21/11/2015	21/11/2015	Same Day	TM-2149031	Tony Molinari	Consumer	Unknown	Camagüey	Camagüey
16997	IN-2012-KB1660058-41136	29/11/2012	30/11/2012	First Class	AW-1084091	Anthony Witt	Consumer	Unknown	Amsterdam	North Holland

Figure 5: Deleted Column

3.2 Aggregation

Problem Identification:

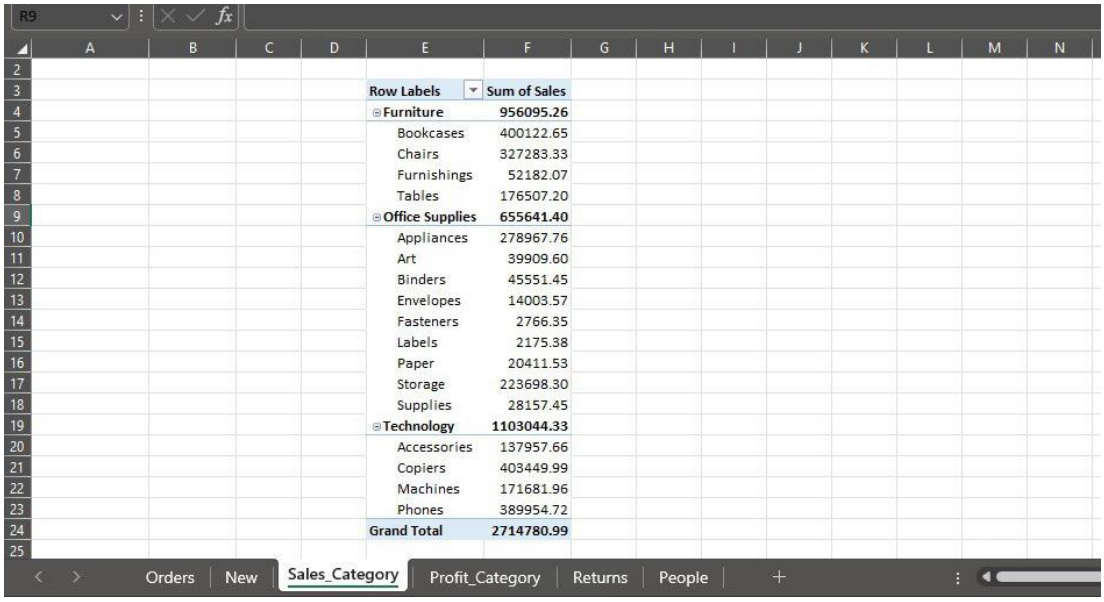
The Global Superstore dataset contains transaction-level data, which makes it difficult to identify overall sales and profit performance across product categories when viewed individually.

Data Reduction Action:

Aggregation was applied to summarise the dataset by grouping products based on Category and Sub-Category, and calculating the total Sales and Profit for each group.

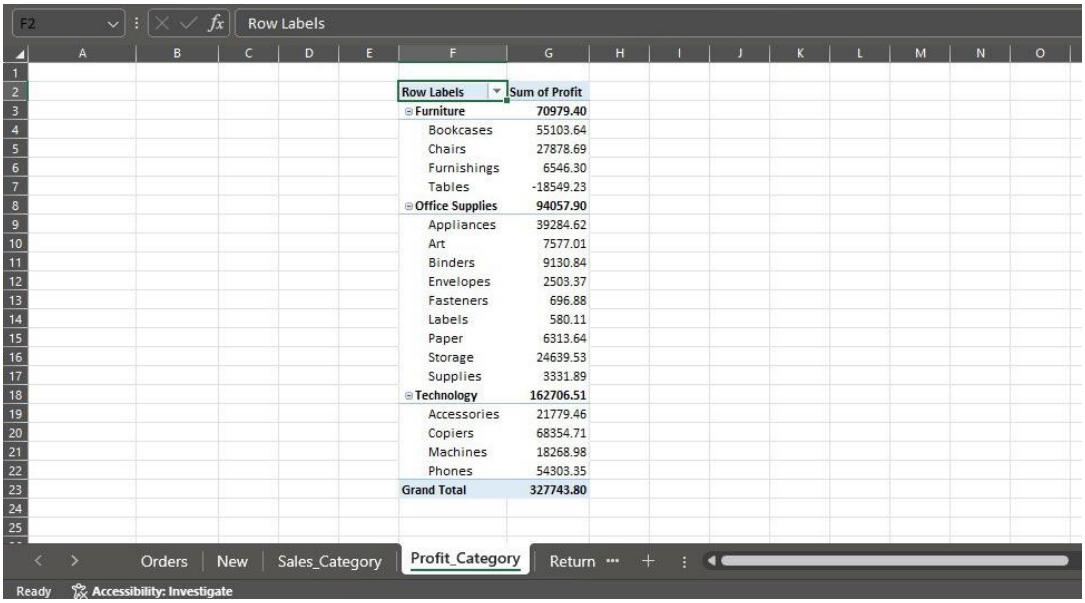
Method Applied:

Aggregation was performed using Pivot Tables in Microsoft Excel. The Category and Sub-Category attributes were placed under Row Labels, while Sales and Profit were set as values and summarized using the Sum function. Separate pivot tables were created to display aggregated sales and aggregated profit.



Row Labels	Sum of Sales
Furniture	956095.26
Bookcases	400122.65
Chairs	327283.33
Furnishings	52182.07
Tables	176507.20
Office Supplies	655641.40
Appliances	278967.76
Art	39909.60
Binders	45551.45
Envelopes	14003.57
Fasteners	2766.35
Labels	2175.38
Paper	20411.53
Storage	223698.30
Supplies	28157.45
Technology	1103044.33
Accessories	137957.66
Copiers	403449.99
Machines	171681.96
Phones	389954.72
Grand Total	2714780.99

Figure 6: Category, Sub-Category, and Sum of Sales



Row Labels	Sum of Profit
Furniture	70979.40
Bookcases	55103.64
Chairs	27878.69
Furnishings	6546.30
Tables	-18549.23
Office Supplies	94057.90
Appliances	39284.62
Art	7577.01
Binders	9130.84
Envelopes	2503.37
Fasteners	696.88
Labels	580.11
Paper	6313.64
Storage	24639.53
Supplies	3331.89
Technology	162706.51
Accessories	21779.46
Copiers	68354.71
Machines	18268.98
Phones	54303.35
Grand Total	327743.80

Figure 7: Category, Sub-Category and Sum of Profit

3.3 Filtering

Problem Identification:

The dataset spans multiple years. Including all years may include irrelevant historical data and make trends less clear.

Data Reduction Action:

The dataset was filtered to include only the recent period (2015–2016) for focused analysis.

Method Applied:

i. The Order Date column was used to filter data in Excel.

ii. Only records from 2015 to 2016 were retained.

iii. The filtered dataset was reviewed to ensure only relevant records remained.

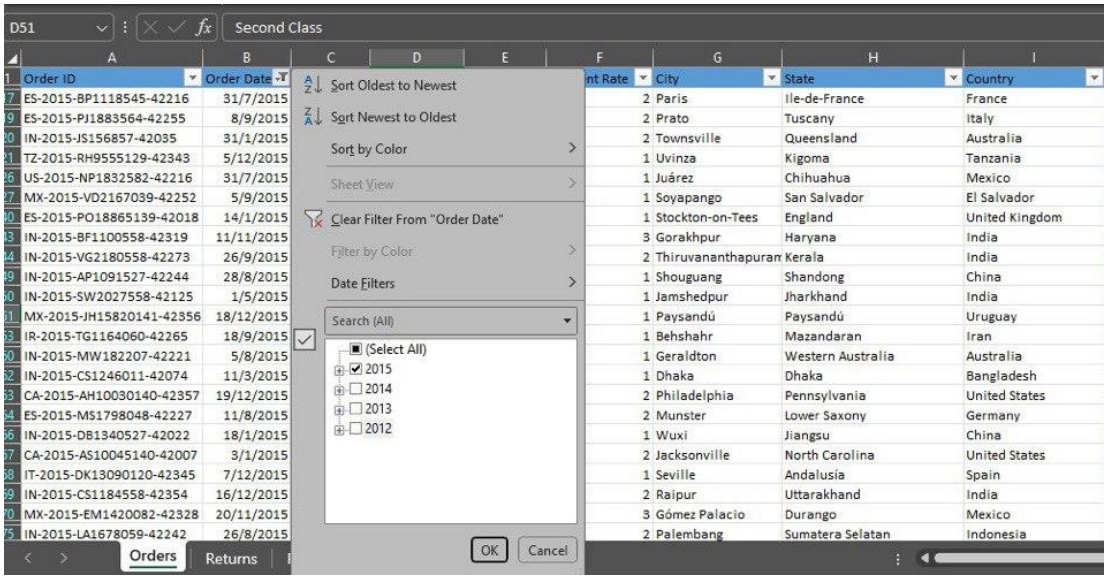


Figure 8: Filtered order date

4.0 Data Transformation

Problem Identification:

Several attributes in the dataset were not in a format suitable for analysis. Categorical attributes, such as Segment and Order Priority, needed to be converted into numeric codes, while numerical attributes like Sales, Profit, Quantity, and Shipping Cost required rounding or standardization to simplify analysis.

Data Transformation Action:

All attributes were transformed into appropriate measurement scales to support analysis:

i. Segment – converted to numeric codes using a mapping table (Nominal).
ii. Order Priority – converted to numeric codes based on priority levels (Ordinal).
iii. Sales and Profit – rounded to the nearest integer (Ratio).
iv. Quantity and Shipping Cost – kept as numeric values (Ratio).

Method Applied:

A mapping table was created for categorical variables, and VLOOKUP was used to assign numeric codes. Decimal values for Sales and Profit were rounded using Excel. The dataset was reviewed to ensure all transformations were applied correctly.



Figure 9: Mapping table for sale column

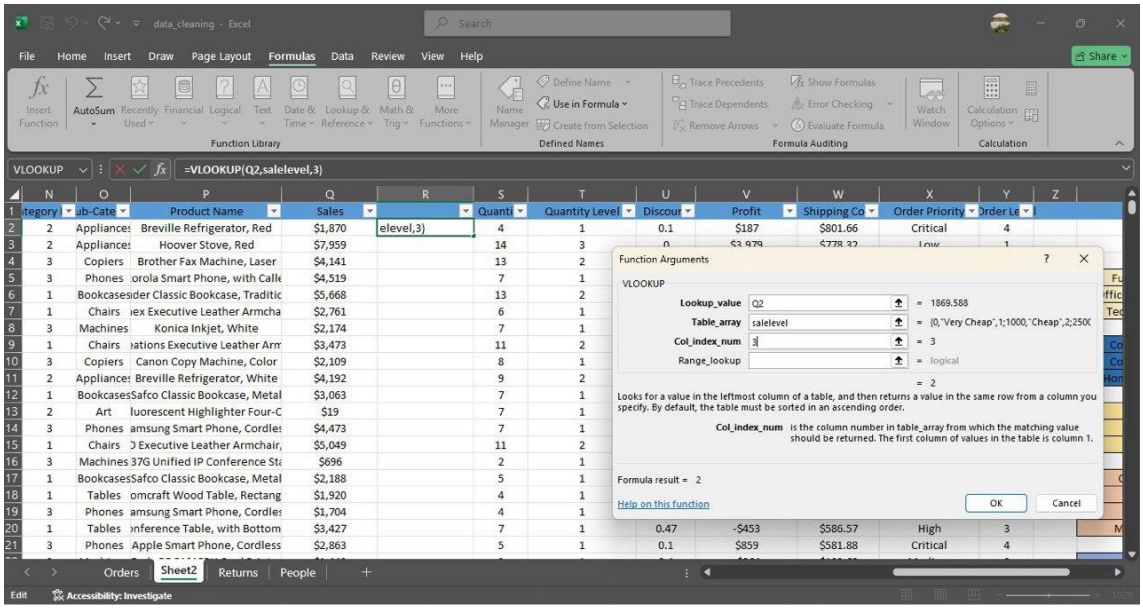


Figure 10: VLOOKUP formulas for the sale column

5.0 Analytics

5.1 Correlation between “Quantity” and “Discount”

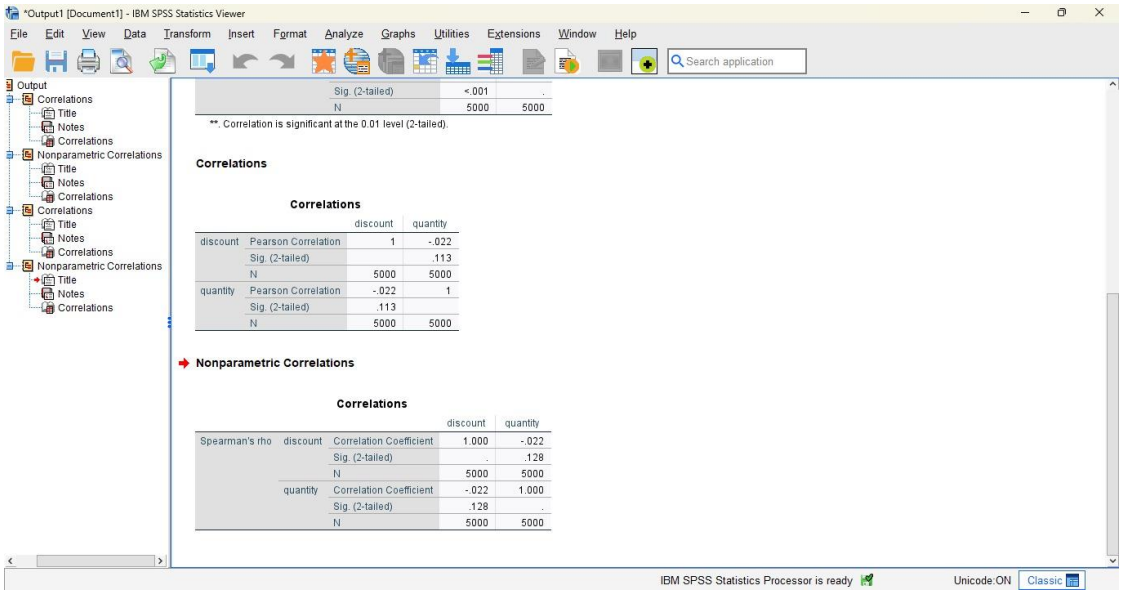


Figure 10: Correlation between “Quantity” and “Discount”

The Spearman’s rho correlation analysis indicates a negative relationship between “Quantity” and “Discount” is very weak (rs = -0.022). The negative value means that the increasing number of “discounts” is related to a small decline in perceived “Quantity”, but the value of the correlation coefficient is very minimal. Moreover, the correlation is not statistically significant which only (p = 0.128). This means that one cannot conclude that there is a significant correlation between “Discount” and “Quantity”. This implies that “discount” levels are not a significant source of perception of “Quantity,” and the observed relationship could not be caused by systematic or practical effect, but rather random variation. As such, the issue of “Discount” cannot be regarded as a valid determinant of variations in “Quantity”.

5.2 Correlation between “Shipping Cost” and “Order Priority”

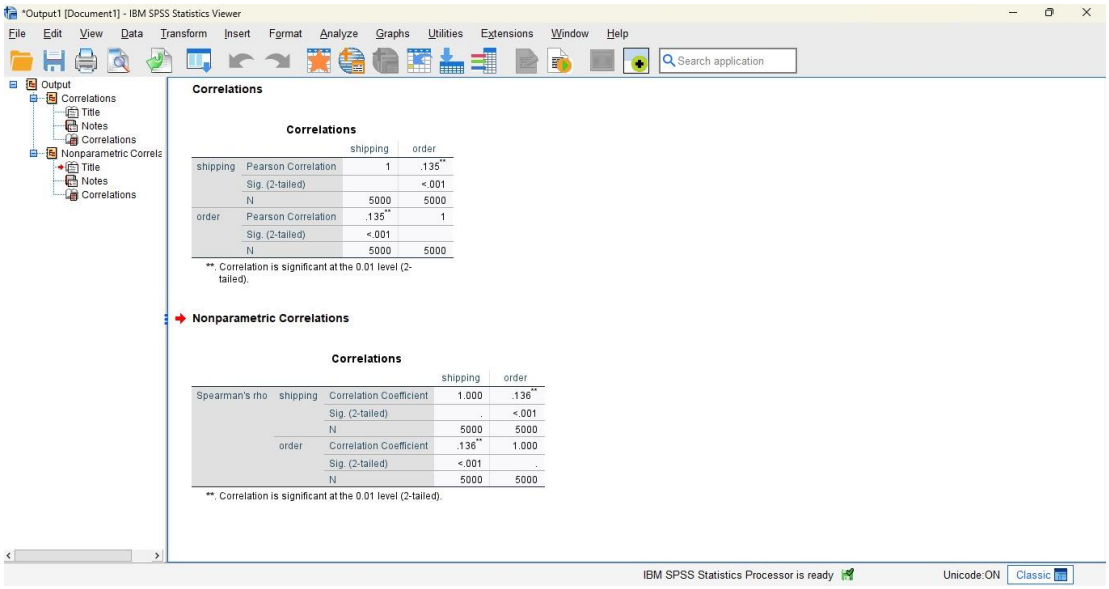


Figure 11: Correlation between “Shipping Cost” and “Order”

In the analysis of Spearman’s rho, the positive correlation between “Shipping Cost” and “Order” is statistically significant (rs= 0.136, p < 0. 01). It means that order values tend to increase slightly as shipping characteristics change. The size of the correlation coefficient, however, indicates that the strength of the relationship is not strong. This means that as much as “Shipping Cost” is connected to “Order”, it has a small contribution to affecting the order patterns. “Shipping cost” is probably not a strong predictor of “Order” behaviour, and there are likely to be other factors beyond shipping that can affect order behaviour.

6.0 Visualization

6.1 Visualization for Ship Mode

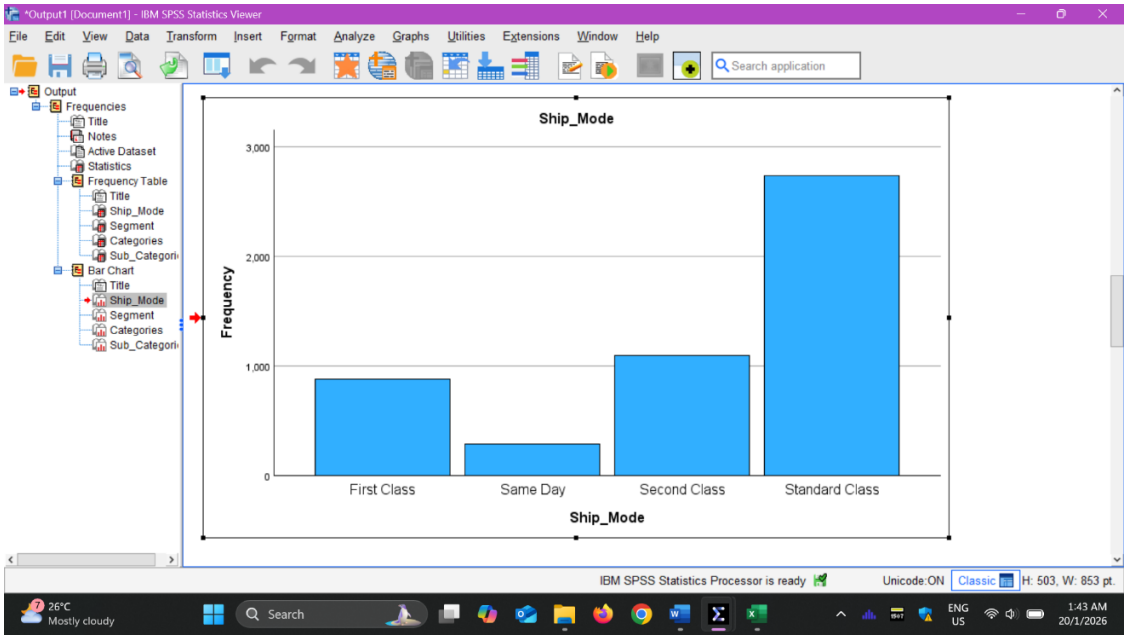


Figure 12: Visualization for Ship Mode

The figure above indicates the frequency of each ship mode in the dataset, which are “First Class”, “Same Day”, “Second Class”, and “Standard Class”. Ship mode is an indicator for the mode of delivery of orders to the customers. The statistics indicate an apparent inclination to “Standard Class” shipping, the most common mode by far, as it is close to 3,000 instances. The next most common are “Second Class” and “First Class” shipping, with the smallest share of the data being the “Same Day” delivery, with less than 500 hits. This distribution indicates that most customers are sensitive to affordable shipping more than speed because the standard, slower, and cost-effective shipping by far exceeds the premium or expedited ones.

6.2 Visualization for Segment

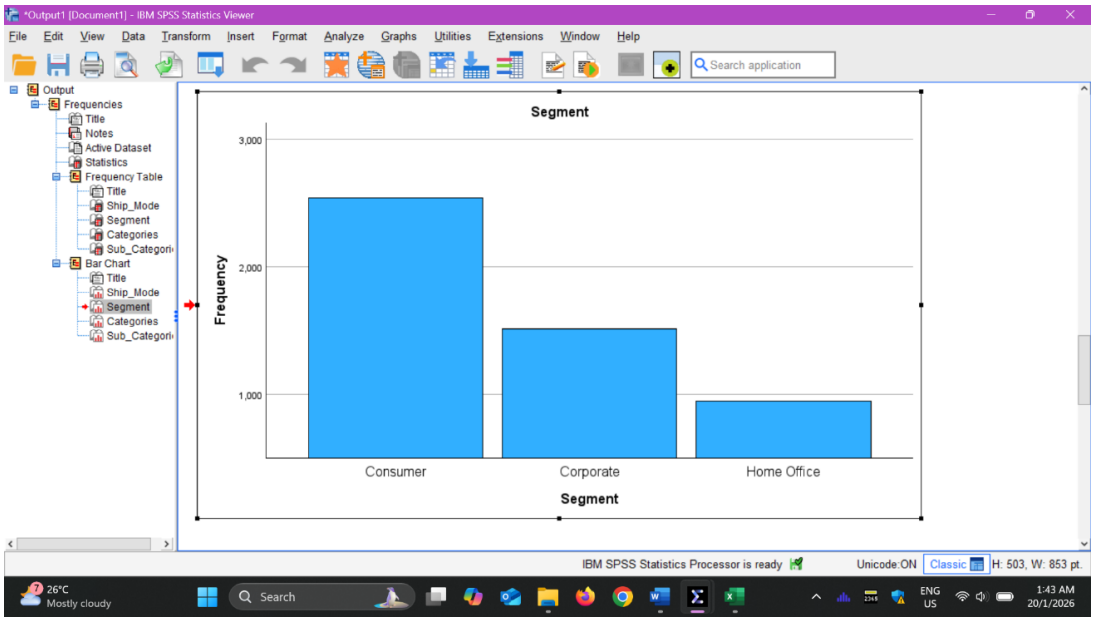


Figure 13: Visualization for Segment

The chart shows the customer distribution in three major market areas, which are “Consumer”, “Corporate”, and “Home Office”. The most noticeable is the “Consumer” segment, which has a frequency count of over 2,500, which means that individual retail shoppers constitute the majority of the dataset. Followed by a moderate presence, the “Corporate” segment and the “Home Office” segment are the least represented compared to the other two segments. This visualization assists in determining the target audience core, in that it is probable that marketing and operational activities are geared towards the individual consumer base rather than the business-to-business dealings, as business is primarily consumer-facing, and resources should likely be prioritized toward the needs and purchasing behaviours of individual shoppers to maximize engagement and revenue.

6.3 Visualization for Categories

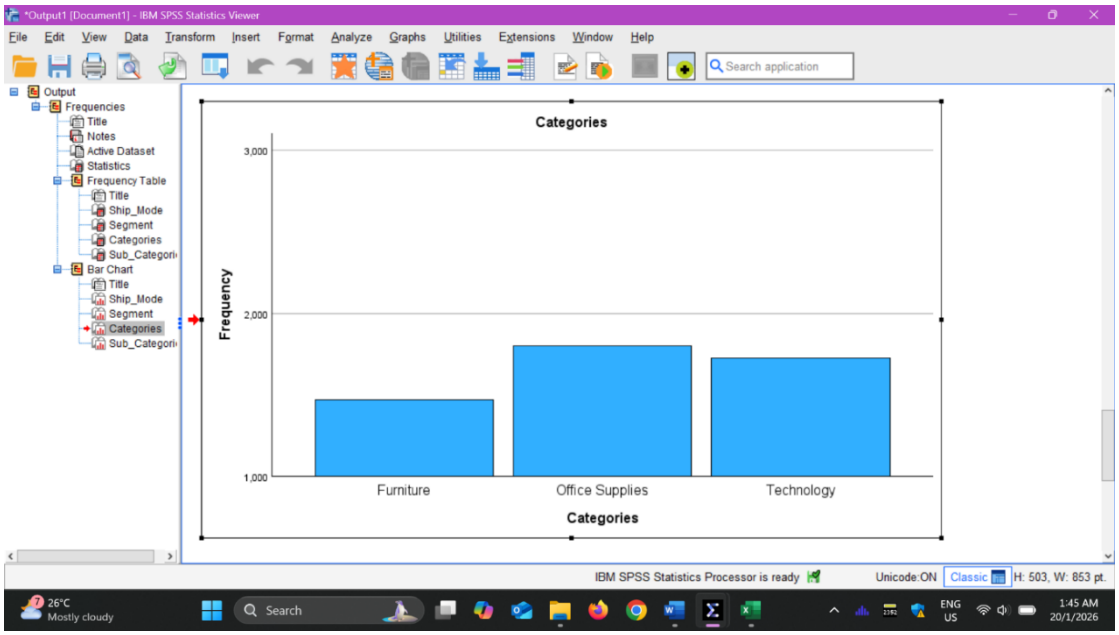


Figure 14: Visualization for Categories

The figure shows a bar graph of the general product lines, namely “Furniture”, “Office Supplies”, and “Technology”. The most frequent category is that of “Office Supplies”, of which the number is nearly close to 2000, indicating that these are the most purchased types of things. “Technology” and “Furniture” have lower frequencies, with “Furniture” being the least prevalent of the three groups. Since “Office Supplies” tend to be less expensive, high turnover products, their preponderance in frequency is not surprising as compared to the more costly and longer life cycle products in the “Furniture” or “Technology” sectors. Understanding this balance helps the organization manage its supply chain by identifying which categories drive consistent foot traffic versus those that generate higher-value, infrequent sales.

6.4 Visualization for Sub-Categories

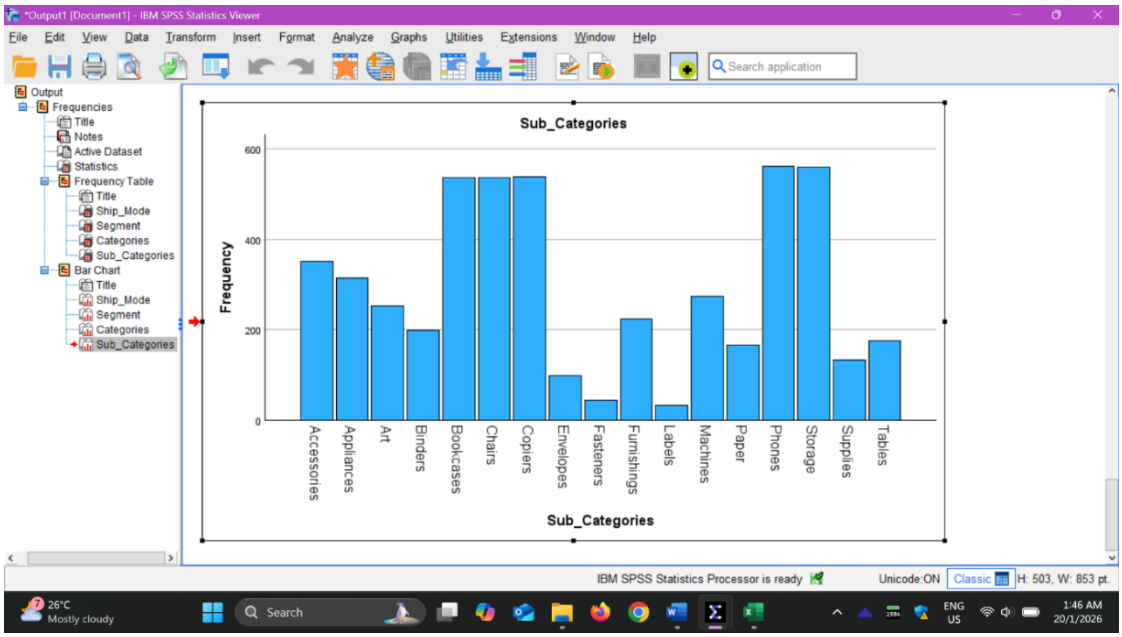


Figure 15: Visualization for Sub-Categories

The chart shows a more in-depth analysis of the data by dividing products into 17 different sub-categories. The horizontal axis lists a diverse range of items from small office consumables like “Labels”, “Fasteners”, and “Art”, to high-value equipment such as “Copiers” and “Machines”. By observing the vertical bars, it becomes immediately apparent that the transaction volume is heavily concentrated in certain areas, especially the most common high-volume products such as “Bookcases”, “Chairs”, “Copiers”, “Phones”, and “Storage”, where it eclipsed the 500-count line. On the other hand, sub-categories such as “Fasteners”, “Labels”, and “Envelopers” have very low frequencies. Such a detail is helpful in the management of inventory, as it singles out exactly what types of products are generating the most transactions in the larger categories considered above. It also provides a much clearer picture of inventory demand and customer purchasing habits.

6.5 Visualization for Product Name

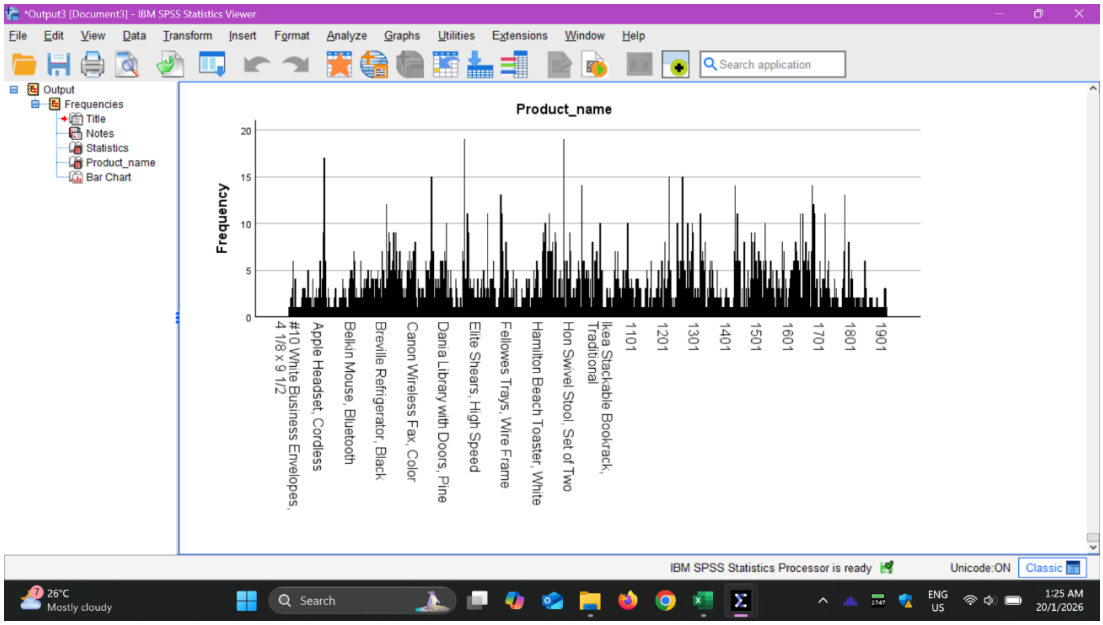


Figure 16: Visualization for Product Names

The picture represents a preliminary effort to visualize the number of all the unique product names in the dataset. Due to the existence of hundreds of unique products, the x-axis is highly saturated, making the labels overlap and useless. Individual product frequencies are very low, with most products having between 1 and 20 frequencies. This chart can be thought of as a snapshot of the data before, and it can be seen that it is quite challenging to visualize high-cardinality categorical data without first applying some filter to the data.

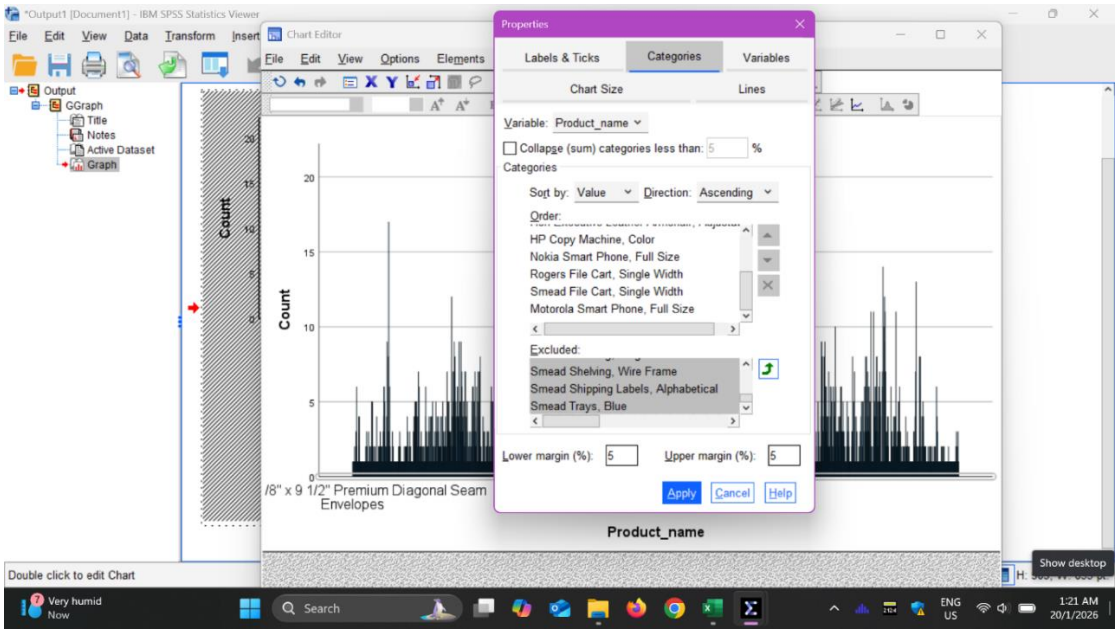


Figure 17: Filtering top 10 Product Names

The picture portrays the process of analysis in the Properties window of the SPSS Chart Editor. The chart is filtered using the categories tab. Several products, including “Smead Shelving”, “Smead Trays”, and many others, are being relocated to the Excluded list in order to make the chart simple. Only the Top 10 most common Product Names are chosen. This is an essential step towards data storytelling because it enables the elimination of any noise and concentrates the attention of the viewer on certain objects of interest, which are the ten best-selling products in the dataset.

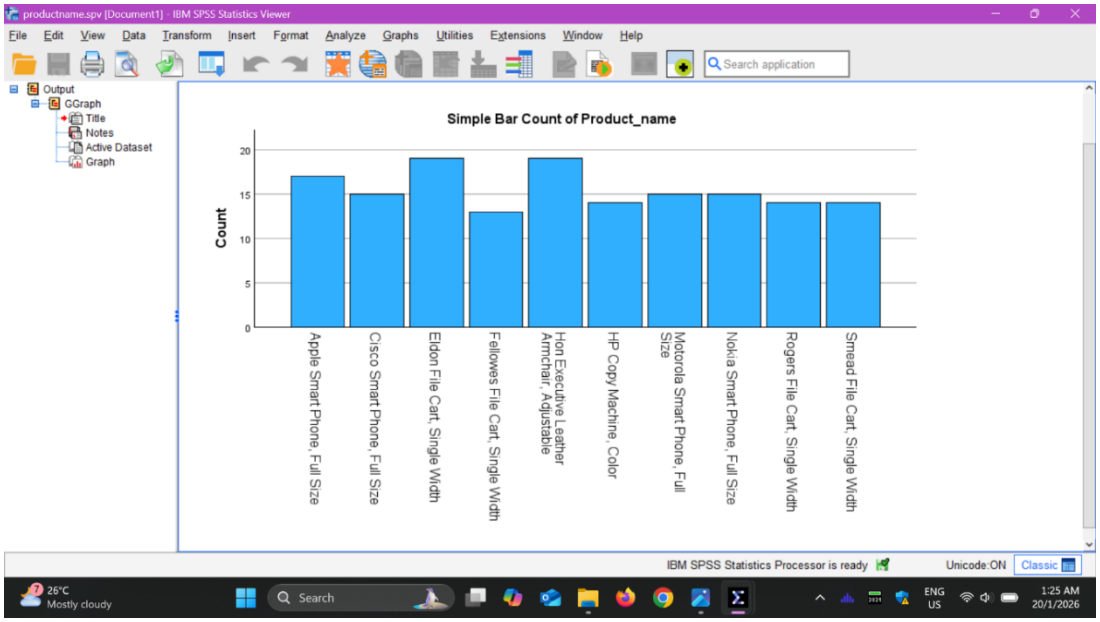


Figure 18: Visualization for Top 10 Product Names

The picture above demonstrates the effective outcome of the filtering process, which presents a clean “Simple Bar Count” with only the top 10 products chosen. The comparison of particular items becomes easy in this sophisticated look. The chosen product, such as the “Eldon File Cart, Single Width” and the “Hon Executive Leather Armchair, Adjustable”, can be observed as the most common in this group, with almost 20 counts each. Other products, such as “Nokia Smart Phone” and “Cisco Smart Phone”, have slightly lower but similar frequencies. The chart is an extremely useful tool used to compare specific high-interest inventory items due to the narrowed scope of the chart.

7.0 Conclusion

The accomplishment of this analytical research on the Global Superstore dataset emphasizes how important systematic data pre-processing and exploratory data analysis are to the field of information analytics. This study effectively negotiated the challenges of raw data management, starting with the discovery and resolution of three different data quality issues during the data cleaning phase. The study stayed concentrated on high-value data chunks thanks to the subsequent usage of data reduction techniques, which improved computational efficiency and the lucidity of the conclusions. Additionally, in order to guarantee that the variables were mathematically compatible for precise measurement, attributes like “Segment”, “Sales”, “Profit”, “Shipping Cost”, and a few more had to be transformed into suitable scales.

Next, through correlation testing in particular, the analytical part of the study gave a deeper knowledge of how different business KPIs interact. The study found important patterns that determine operational costs and consumer behaviour by examining the connections between “Order Quantity” and “Discount” as well as the effect of “Order Priority" on “Shipping Costs”. Multi-dimensional representations of “Ship Modes”, “Product Categories”, “Sub-Categories”, Segments”, and “Product Name” further enhanced these quantitative findings by converting abstract data points into useful business insight using SPSS Statistics. In the end, this assignment shows that the accuracy of the initial data preparation greatly influences the integrity of information analytics, offering a strong framework for evidence-based decision-making in a worldwide commercial setting.